

DA3452 Speech and Language Technology

Module 3 Test

Learning and Generating Sequences

J&M Chapter 6 and classroom material through Lecture 17

Duration: 60 minutes

Maximum marks: 60

Name:	Roll No.:	Date:
-------	-----------	-------

Instructions

- Section A: 30 MCQs, 1 mark each. Section B: 10 MCQs and 5 fill-in questions, 2 marks each. All questions are compulsory.
- Each MCQ has exactly one best answer. There is no negative marking. Each fill-in has two blanks, worth 1 mark each.
- Use the course gate convention: a larger GRU update gate puts more weight on the candidate. Gate values of 0 or 1 denote idealized limiting cases. $\odot$ denotes elementwise multiplication.
- Tensor shapes omit the batch dimension unless stated. Equivalent exact answers or decimals correct to three places are accepted. No lengthy gradient calculations are required.
- Scope ends at Lecture 17, including fixed-context encoder-decoder models. Attention and Transformers are excluded.

SECTION A: APPLIED FOUNDATIONS

Questions 1–30 | 1 mark each | 30 marks

1. A classifier averages context-independent word embeddings before applying a multilayer network. The sentences “dog bites man and “man bites dog receive different gold labels. What limitation follows?
 - A. A deeper downstream network can recover the word order from the same average.
 - B. The averaged inputs are identical, so this deterministic classifier cannot give them different predictions.
 - C. The averages differ because each word occurs at a different position.
 - D. Learning the embedding table will give the two averages different values.
2. A binary classifier stacks several affine layers, with no activation between them, and ends with a sigmoid. What does adding more such hidden layers change about its decision boundary in the original input space?
 - A. Each extra layer introduces an additional nonlinear bend in the boundary.
 - B. The boundary remains affine, although the parameterization becomes deeper.
 - C. The boundary becomes nonlinear whenever the hidden layers have different widths.
 - D. The final sigmoid makes the boundary capable of separating XOR.
3. A network produces logits $\mathbf{z}$. Before softmax, a programmer adds the same constant to every component. What happens to the output probabilities?
 - A. They move closer to a uniform distribution.
 - B. The largest probability increases while the others decrease.
 - C. They remain unchanged because the common exponential factor cancels.
 - D. They all increase by the same amount.
4. On one example, two classifiers both predict the correct class. They assign that class probabilities 0.6 and 0.9. Which comparison is correct?
 - A. The second has higher accuracy, but their cross-entropy is equal.
 - B. The first has lower cross-entropy because it is less confident.
 - C. Their accuracy on this example is equal, but the second has lower cross-entropy.
 - D. Their accuracy and cross-entropy are both equal.

5. An embedding vector feeds a differentiable layer whose weight matrix is frozen. The embedding table remains trainable, and the computation is not detached. Can the loss still train the embedding?
- A. No; a frozen matrix always blocks the backward path through it.
 - B. Only the embedding bias can change; its coordinates cannot.
 - C. Only if the fixed matrix is updated once per epoch.
 - D. Yes; derivatives can pass through the fixed layer to the embedding.
6. Two hidden units have identical incoming weights and biases, identical outgoing weights, and the same activation. Under deterministic gradient descent with no dropout, what problem can this symmetry cause?
- A. The output softmax assigns a different training example to each unit.
 - B. The symmetry disappears automatically after the first backward pass.
 - C. The symmetry forces both units to learn opposite features.
 - D. They can keep receiving identical updates and fail to learn distinct features.
7. A student chooses hidden-layer width, initializes the weights, and trains with backpropagation and gradient descent. Which statement correctly describes what is learned in this setup?
- A. Backpropagation learns the width and leaves the initialized weights fixed.
 - B. The weights are optimized within the chosen architecture; its width does not change automatically.
 - C. Both width and weights are automatically optimized by the same gradient update.
 - D. Only the final layer can learn because earlier computations were specified by the designer.
8. During training with dropout, a hidden unit is masked on one mini-batch. Which interpretation is correct?
- A. The unit receives a new random weight vector before the next mini-batch.
 - B. Its output is temporarily zeroed; its parameters remain available on later mini-batches.
 - C. Its output is replaced by the mean output of all unmasked units.
 - D. The unit and all its parameters are permanently deleted.
9. A vanilla RNN is applied to a sequence twice as long, with the same input and hidden dimensions. Which change is necessary?
- A. More recurrent applications, but no additional distinct recurrent weights.
 - B. Twice as many distinct recurrent matrices, one set for each half.
 - C. Twice the hidden width so that every earlier token has its own coordinate.
 - D. A larger recurrent matrix, but the same number of recurrent applications.
10. The same word appears after two different prefixes. A model uses one context-independent embedding lookup followed by a left-to-right RNN. Which quantities must be identical at the two occurrences?
- A. Neither must match because every position has its own embedding parameters.
 - B. The hidden states must match; the lookup embeddings need not.
 - C. The lookup embeddings must match; the recurrent hidden states need not.
 - D. Both the lookup embeddings and hidden states must match.
11. For every example in a sequence-classification task, an RNN reads the full sequence but incurs a loss only at the final state. Under full BPTT, can an early state affect the parameter gradient?
- A. No; each time step needs its own attached loss to receive credit.
 - B. Only if the classifier weights are different at every time step.
 - C. Yes; the final loss can propagate through the recurrent path to earlier computations.
 - D. Only the input matrix can receive early contributions; the recurrent matrix cannot.

12. A recurrent matrix is shared at all time steps. A programmer makes independent trainable copies for the unrolled steps and updates them separately. What has changed?
- A. The model no longer enforces one shared transition rule across time.
 - B. The hidden-state values must now be identical at all time steps.
 - C. The model is still tied because the copies started with equal values.
 - D. Only the numerical storage has changed; the models remain equivalent after any updates.
13. An RNN carries its hidden-state value from one training chunk to the next but detaches that value from the earlier computation graph. What is preserved across the boundary?
- A. Forward information in the state, but not backward credit through the detached history.
 - B. Both forward information and the entire earlier backward path.
 - C. Neither forward information nor the carried numerical state.
 - D. Backward credit through the entire prefix, but no forward information.
14. An RNN frequently produces very large gradient norms. After norm clipping is added, updates stabilize, but learning distant dependencies remains poor. Why is this possible?
- A. A stable update guarantees reliable long-range credit assignment.
 - B. Clipping limits large updates but does not restore gradients that have already vanished.
 - C. Clipping removes the recurrent state, so all long-range information is erased.
 - D. Clipping increases all small gradients to the clipping threshold.
15. The recurrent matrix has moderate entries, yet gradients through a long tanh RNN become very small. Which observation could help explain this?
- A. Moderate recurrent entries guarantee that every recurrent Jacobian has norm one.
 - B. A hidden state with fixed width cannot have small gradients.
 - C. Parameter sharing prevents local activation derivatives from affecting BPTT.
 - D. Many hidden units operate in the saturated part of tanh, giving small local derivatives.
16. At deployment, a sentiment RNN processes unrelated reviews in random order. Its hidden state is accidentally carried from each review into the next. What unintended dependence can result?
- A. A review prediction may depend on which unrelated review was processed immediately before it.
 - B. The recurrent weights become different for each review length.
 - C. The model automatically becomes bidirectional within each review.
 - D. The embedding dimension grows after each review.
17. A fixed-window language model and an RNN both have hidden width d . Which statement about accessible history is accurate?
- A. Equal hidden width gives both models exactly the same context window.
 - B. The RNN can always reconstruct its entire input history exactly.
 - C. A wider hidden layer lets the fixed-window model directly observe omitted earlier tokens.
 - D. The fixed-window input has a hard cutoff; the RNN can carry older information, without guaranteeing that it will retain it.
18. A full GRU must largely ignore the previous state when constructing a candidate, but mostly retain that previous state in the final blend. Under the course convention, which setting fits?
- A. Reset gate near one; update gate near zero.
 - B. Reset gate near zero; update gate near one.
 - C. Reset gate near one; update gate near one.
 - D. Reset gate near zero; update gate near zero.

19. A GRU uses $\Gamma_{u,t} = 1$ in one coordinate, but its candidate equals the old hidden-state value in that coordinate. What follows from the blend equation?
- A. The numerical state is unchanged even though the candidate receives all the blend weight.
 - B. The state becomes twice its previous value.
 - C. The state must change because the update gate is one.
 - D. The coordinate becomes zero because no old-state weight remains.
20. A programmer replaces a d -component sigmoid GRU update gate with a softmax over its d components, where $d > 1$. Which useful behaviour becomes impossible?
- A. Having every coordinate's update gate simultaneously close to one.
 - B. Assigning different update weights to different state coordinates.
 - C. Obtaining a gate vector with the same length as the hidden state.
 - D. Computing the gates from both the current input and previous state.
21. A GRU's gate matrices are shared across time. Must its gate values therefore be constant throughout a sentence?
- A. Yes; only the candidate can depend on the current token.
 - B. No; the same matrices process changing inputs and hidden states.
 - C. Yes; a new gate value requires a new set of trainable matrices.
 - D. No; the matrices are retrained separately before every forward step.
22. In one LSTM coordinate, the forget gate is one, the update gate is zero, and the output gate is zero. Which statement is correct?
- A. The candidate overwrites the memory but is hidden from the output.
 - B. Both the cell memory and visible hidden coordinate are erased.
 - C. The cell memory is retained while the visible hidden coordinate is zero.
 - D. The old memory is copied directly into the visible hidden coordinate.
23. An LSTM computes its forget, update, and output gates from the same concatenated input. A programmer also ties all three gate matrices and biases to identical values. What restriction does this introduce?
- A. The three sigmoid gate vectors become identical for a given input, removing their independent control.
 - B. None; a common input already forces the three standard gates to have identical values.
 - C. The three gates can still differ because they are used at different locations after computation.
 - D. Only the gate-vector lengths change; their numerical values are unaffected.
24. Assume the GRU candidate and old hidden coordinate both lie in $[-1, 1]$. Compare its standard blend with the LSTM cell update. Which statement is correct?
- A. The GRU blend can exceed both inputs because its gate is learned.
 - B. Both updates must remain convex combinations of old and candidate memory.
 - C. The GRU blend stays between its two inputs; an LSTM cell can exceed 1 through additive accumulation.
 - D. The LSTM cell is bounded by tanh because its visible state uses tanh.
25. An LSTM is given a task with a distant dependency. Which claim about its additive memory path is most accurate?
- A. Its existence makes every total derivative exactly one.
 - B. It removes all indirect gradient paths through hidden states and gates.
 - C. It provides a route that can preserve credit, but learned gate settings still matter.
 - D. It guarantees exact retention of every earlier input.

- 26.** A two-layer left-to-right RNN is proposed for immediate streaming predictions. What does adding the second recurrent layer provide?
- A.** Right-context information because the upper layer runs backward.
 - B.** The same effect as doubling the sequence length.
 - C.** A separate trainable parameter set for every input position.
 - D.** A deeper transformation of the available prefix, without access to unseen future inputs.
- 27.** An offline bidirectional RNN tagger uses right context. The same task must now emit irrevocable tags as each token arrives, with no delay. Which limitation matters?
- A.** Bidirectional states cannot be used for token-level predictions even offline.
 - B.** The forward states also require the entire future sentence.
 - C.** The original backward states require future tokens that are unavailable at the emission time.
 - D.** Replacing the recurrent cells with LSTMs supplies the missing future tokens.
- 28.** A fixed-context encoder-decoder uses the same source vector $\mathbf{c}_{\text{ctx}}$ at every decoder step. Why can its next-token distributions still differ from step to step?
- A.** The source vector is necessarily a new trainable parameter at each step.
 - B.** A fixed source vector forces an identical distribution until the encoder is rerun.
 - C.** The output projection must have different weights at each step.
 - D.** The previous target input and recurrent decoder state can change.
- 29.** During teacher forcing, the decoder's prediction at step 2 is wrong. When forming the state for step 3, which input is normally used for the previous target token?
- A.** The gold token from target position 3; it is the label currently being predicted.
 - B.** The wrong predicted token from step 2; the recurrent state is reset.
 - C.** The correct source token at position 2; target tokens are never decoder inputs.
 - D.** The gold token from target position 2; the decoder recurrent state still carries forward.
- 30.** A recurrent encoder-decoder translates a six-token source into an eight-token target. Which feature permits this without learning a different model for this length pair?
- A.** The decoder has its own timeline and generates an end-of-sequence token.
 - B.** The decoder must change its recurrent matrix when its seventh step begins.
 - C.** The encoder's hidden width must equal the target length.
 - D.** Every source state must be paired with exactly one target token.

SECTION B: ADVANCED ANALYSIS

Questions 31–45 | 2 marks each | 30 marks

31. A ReLU hidden unit has a strictly negative preactivation on every training example. Its output is zero on all of them. With ordinary backpropagation, no regularization, and no other path using its incoming weights, what happens to those weights' loss gradients?

- A. They equal the output-layer gradients because zero activation is an identity map.
- B. They are zero on these examples because the local ReLU derivative blocks this path.
- C. They must be positive because a negative preactivation needs to be increased.
- D. They become nonzero merely by adding more identical training examples.

32. A fixed-window neural LM has vocabulary size 20 and embedding width 4. It looks up three tokens, concatenates their column embeddings into $\mathbf{x}$, and computes $\mathbf{h} = \tanh(W\mathbf{x} + \mathbf{b})$ with hidden width 5. The embedding table has one row per vocabulary item.

The embedding-table shape is _____. The shape of W is _____.

33. A trainable embedding table is used only for input lookup. One token occurs twice in a training example, in different contexts. The loss gradients reaching its two lookup outputs differ. Under ordinary backpropagation, how is that token's table row trained?

- A. The two gradient contributions are added into the same row before its update.
- B. The table permanently creates two different rows for the two contexts.
- C. Only the later occurrence contributes because it is closer to the final loss.
- D. The larger gradient replaces the smaller one to avoid double counting.

34. An RNN processes $x_1, \dots, x_8$ in two chunks. It carries the value of $\mathbf{h}_4$ into the second chunk but detaches its history. A loss is attached only after x_8 . Consider this single backward pass. Which statement is correct?

- A. The first chunk receives the full loss gradient because its state value was retained.
- B. The loss may depend on information from x_1 , but no gradient reaches the first chunk through the detached state.
- C. Detachment removes every influence of x_1 from the forward prediction.
- D. The shared recurrent matrix receives no gradient anywhere in either chunk.

35. Global norm clipping applies $\mathbf{g}_{\text{clip}} = \mathbf{g} \min(1, \tau / \|\mathbf{g}\|_2)$. Let $\mathbf{g} = (6, 8)$, whose norm is 10, and let $\tau = 5$.

The first component of $\mathbf{g}_{\text{clip}}$ is _____. The second component is _____.

36. For illustration, a scalar recurrence is $h_t = ah_{t-1} + x_t$ with $h_0 = 0$. After two inputs, $h_2 = ax_1 + x_2$. For the orders (p, q) and (q, p) , with $p \neq q$, which conclusion follows?

- A. The final states differ for every a because the computation is recurrent.
- B. The final states coincide for every a because the same weights are shared.
- C. The final states coincide only when $a = 0$, because the previous state is ignored.
- D. If $a = 1$, the two final states coincide; if $a \neq 1$, they differ.

37. For a fixed sequence of length $T > 1$, two training runs start with identical RNN parameters and states. Run A uses $\sum_t \ell_t$; run B uses $T^{-1} \sum_t \ell_t$. Both use plain SGD with the same learning rate, without clipping or regularization. How do their first parameter updates compare?

- A. Run B changes only the output-layer update; recurrent updates remain unchanged.
- B. Run B has the same update direction and $1/T$ of the magnitude, when the gradient is nonzero.
- C. Run B generally reverses the update direction because its losses are averaged.
- D. The updates are identical because parameters are shared across time.

38. The course GRU resets old-state coordinates before a matrix mixes them. A programmer gates the mixed old-state contribution instead. For a general dense matrix and unequal reset values, are these operations interchangeable?

- A. Not generally; suppressing source coordinates differs from suppressing mixed destination coordinates.
- B. Yes; matching vector lengths make the two expressions identical.
- C. Not generally; moving the gate would force the candidate to ignore the current input.
- D. Yes; elementwise gating always commutes with matrix multiplication.

39. Use the course GRU blend, coordinate by coordinate:

$$\mathbf{h}_t = (\mathbf{1} - \mathbf{\Gamma}_{u,t}) \odot \mathbf{h}_{t-1} + \mathbf{\Gamma}_{u,t} \odot \tilde{\mathbf{h}}_t.$$

Let $\mathbf{h}_{t-1} = (0.6, -0.4)$, $\tilde{\mathbf{h}}_t = (-0.2, 0.8)$, and $\mathbf{\Gamma}_{u,t} = (0.25, 0.5)$.

The first component of $\mathbf{h}_t$ is _____. The second is _____.

40. For one LSTM coordinate, use

$$c_t = \Gamma_{f,t}c_{t-1} + \Gamma_{u,t}\tilde{c}_t, \quad h_t = \Gamma_{o,t} \tanh(c_t).$$

Let $c_{t-1} = 0.6$, $\tilde{c}_t = 0.8$, $\Gamma_{f,t} = \Gamma_{u,t} = 0.9$, and $\Gamma_{o,t} = 0$.

The new cell value c_t is _____. The visible value h_t is _____.

41. An LSTM has forget gates close to one for several steps. A student claims that the *total* derivative of a later cell state with respect to an earlier one must therefore equal one. What is the best correction?

- A. The conclusion is exact because additive updates have no other backward paths.
- B. The direct memory-path factor can be near one, while indirect paths through states, gates, and candidates also contribute.
- C. The total derivative is determined only by the final output gate.
- D. The direct path is always zero because the forget gate is itself learned.

42. A stacked bidirectional RNN concatenates first-layer states of width 3 from each direction. Each second-layer direction receives this concatenation and has hidden width 4. A token classifier receives the concatenated second-layer states.

Each second-layer direction receives _____ features; the classifier receives _____.

43. Two source sentences give a deterministic decoder identical context vectors and initial states. It is fed the same target prefix and has no other access to the source. What follows at the next step?

- A. The decoder can reconstruct a different missing source detail from the identical context in each case.
- B. The distributions must differ because the source sentences are different.
- C. It must produce the same next-token distribution for both sources under these conditions.
- D. A larger decoder can distinguish the sources even though all its inputs coincide.

44. A jointly trained encoder-decoder supplies its fixed source context at every decoder step. Does teacher forcing make the encoder irrelevant to the loss gradient?

- A. No, but only the first decoder loss can reach a context reused at later steps.
- B. Yes; gold target tokens remove all dependence on the source context.
- C. No; decoder losses can still backpropagate through the context to the encoder.
- D. Yes; the encoder can receive a gradient only through incorrectly generated tokens.

45. A decoder scores well with gold previous tokens, yet degrades after an early mistake in free-running generation. Which explanation best connects these observations?

- A. The recurrent weights are updated automatically after each generated token.
- B. An early wrong token changes the source sentence and forces the encoder to reread it.
- C. Teacher forcing stops the decoder from learning any conditional probabilities.
- D. Generation conditions later steps on the model's own imperfect prefix, which the gold-prefix evaluation did not test.